

Additional file 1: Supplementary methods and figures

Supplementary methods

Study population and phenotype definition

The source export contained 2,548 records and 2,279 unique patients. After completeness-based patient-level consolidation, 2,010 patients had one source record and 269 had two source records. The extract did not establish that these records represented distinct admissions or identify a uniform index angiography episode. Among 1,944 patients with discharge-diagnosis text, 456 were definite AMI, 1,372 were definite non-AMI CAD, 5 had an ambiguous MI mention, and 111 did not meet the CAD/AMI target; 335 additional records lacked a discharge diagnosis. Required absolute neutrophil, lymphocyte, monocyte, and platelet data were available in 453 AMI and 1,367 non-AMI CAD patients. AMI was classified from discharge-diagnosis text using explicit current acute/subacute MI wording or unambiguous STEMI/NSTEMI/subtype descriptions. Old/prior MI, history of MI, prior PCI, CAD, stable angina, and unstable angina alone were not AMI-positive. Troponin and CK-MB were not used to reclassify the full cohort: assay platforms, ULNs, serial changes, and index timing were not consistently available. No independent cardiologist, ECG, imaging, or uniform biomarker adjudication was performed for all patients. The phenotype is therefore discharge-diagnosis-text based and is not formal Fourth Universal Definition adjudication. Mapping details and counts are in Additional file 2, Table S2.

Predictor preprocessing and model development

Core-7 included absolute neutrophil, lymphocyte, and monocyte counts, platelet count, MPV, RDW, and hemoglobin; fibrinogen was added only in the Enhanced model. Transformations and standardization were estimated within training folds. A completeness-based rule selected

one record per patient by prioritizing non-empty discharge diagnosis, completeness of the four absolute counts, a parsable WBC/CBC timestamp, overall non-missingness, and original source-row order as tie-break. Because encounter-level linkage and angiography/diagnosis timestamps were not sufficiently complete, selected CBC measurements cannot be assumed to be first, admission-first, pre-angiography, pre-diagnostic, or pre-treatment. All 1,820 measurements remain timing-unverified relative to the index angiography hospitalization.

Internal validation and sensitivity analyses

Primary internal validation used stratified outer five-fold cross-validation repeated 10 times and five-fold tuning within outer training sets. Each patient received one held-out probability per repeat; the arithmetic mean of the 10 held-out probabilities was used for patient-level evaluation. Models shared outer partitions for paired comparisons where applicable. Imputation, transformation, centering/scaling, and tuning were confined to training folds. Confidence intervals were based on 1,000 patient-level bootstrap resamples of the fixed mean out-of-fold predictions and outcomes, without refitting; intervals therefore condition on those predictions and omit full model-development uncertainty. Clinical incremental analyses used age, sex, hypertension, and diabetes; smoking was unavailable. Full definitions and results are in Additional file 2, Tables S6-S8.

An exploratory deidentified date-axis sensitivity analysis used the prespecified development, buffer, and later groups: development N=1,001 (AMI=196), buffer N=271 (AMI=79), and later N=548 (AMI=178). The earlier group was used for model development and the later group was evaluated as a fixed held-out date-axis group. The selected laboratory dates could not be uniformly linked to the index angiography hospitalization; this analysis is not formal temporal validation and does not establish transportability. A separate conservative +/-365-day sensitivity result is reported with its source confidence intervals in Table S11.

Exploratory simplified score

A point-based simplification was explored during prior model development but showed inferior calibration and is not the model evaluated in the current revised results. It is retained only for transparency in Additional file 2, Table S12, and is not proposed for clinical use.

Conventional indices and PIV comparison

All seven conventional indices were assessed using the existing repeated nested cross-validation outputs. PIV was the prespecified primary comparator; NLR, PLR, MLR, SII, SIRI, and HRR are supplementary benchmarks. Estimates and 95% confidence intervals are listed in Additional file 2, Table S4; no additional primary hypothesis tests were introduced.

Clinical and fibrinogen complete-case comparisons

Fibrinogen was missing in 115/1,820 patients (6.32%), 27/453 AMI patients (5.96%), and 88/1,367 non-AMI CAD patients (6.44%). Core-7 and Enhanced complete-case models were compared in the identical 1,705 patients (426 AMI, 1,279 non-AMI CAD) and identical validation partitions; their AUCs and paired difference are reported in Tables S9-S10. In this same 1,705-patient sample, AUCs were 0.546 (95% CI 0.515-0.574) for Clinical, 0.717 (0.689-0.745) for Clinical+PIV, 0.730 (0.701-0.755) for Clinical+Core-7, and 0.741 (0.712-0.770) for Clinical+Enhanced. Table S8 also reports Brier scores, calibration statistics, and paired Δ AUC estimates.

Calibration and decision-curve analysis

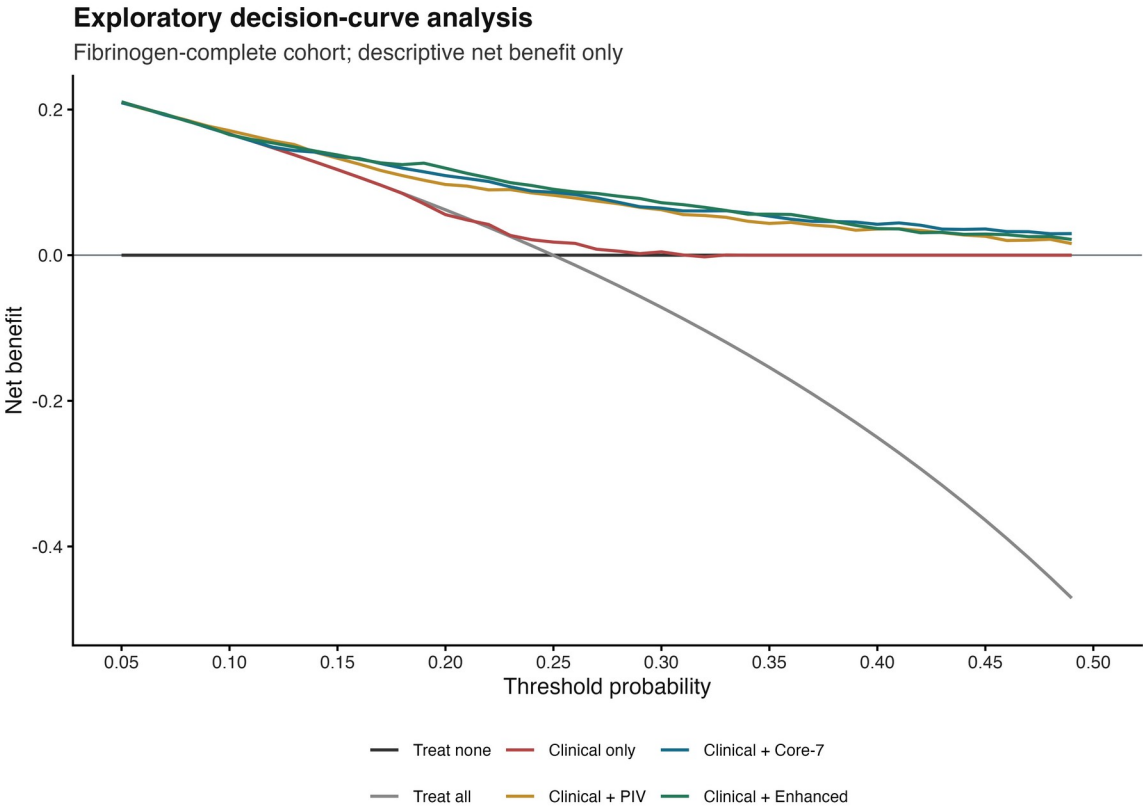
Calibration plots use the same patient-level mean 5-fold-by-10-repeat held-out probabilities underlying the reported calibration statistics. Ten quantile groups are shown where supported by the analysis source; group-specific sample sizes and predicted-probability summaries are provided in Additional file 2, Table S6. Exploratory decision-curve analysis used the four clinical models plus treat-all and treat-none references over thresholds 0.05-0.50. It is

73 descriptive and does not establish clinical utility, treatment benefit, or a preferred operating
74 threshold.

75 **Date-axis and phenotype sensitivity**

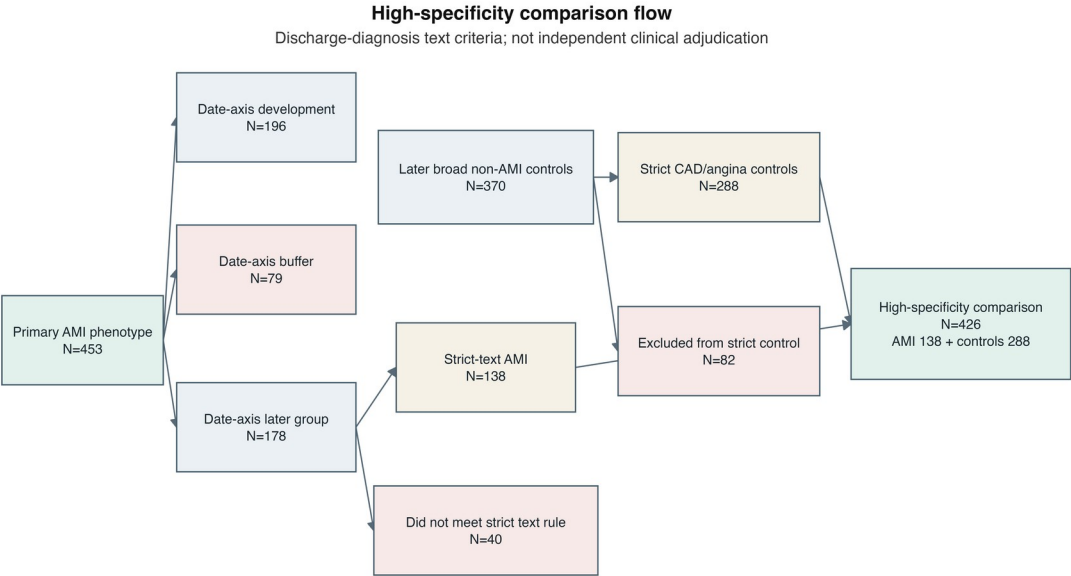
76 In the later date-axis group, Core-7 AUC was 0.701, PIV 0.699, and Enhanced 0.719; Core-7
77 minus PIV Δ AUC was 0.002 (95% CI -0.021-0.025). The advantage attenuated and the interval
78 crossed zero. The analysis is not formal temporal validation because available dates were not
79 uniformly linked to the index angiography hospitalization.

80 The high-specificity comparison used 138 definite AMI text cases and 288 strict CAD/angina
81 controls. Of 178 AMI records in the prespecified later date-axis group, 138 met the stricter text
82 rule and 40 did not; the latter are not adjudicated misclassifications. The stricter rule required
83 an MI keyword plus an acute marker, excluded subacute and uncertain wording, and required
84 a STEMI/NSTEMI/ST-segment subtype or explicit acute-MI phrase/site. The criteria are based
85 on discharge information and are not a prospective baseline rule.



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88 Figure S1. Exploratory decision-curve analysis in the fibrinogen-complete sample (N=1,705; AMI=426).
89 Thresholds span 0.05-0.50; curves are descriptive and do not establish clinical utility.



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Figure S2. High-specificity phenotype comparison flow. The 138 strict-text AMI cases and 288 strict CAD/angina controls were classified from discharge-diagnosis text; this was not independent clinical adjudication.

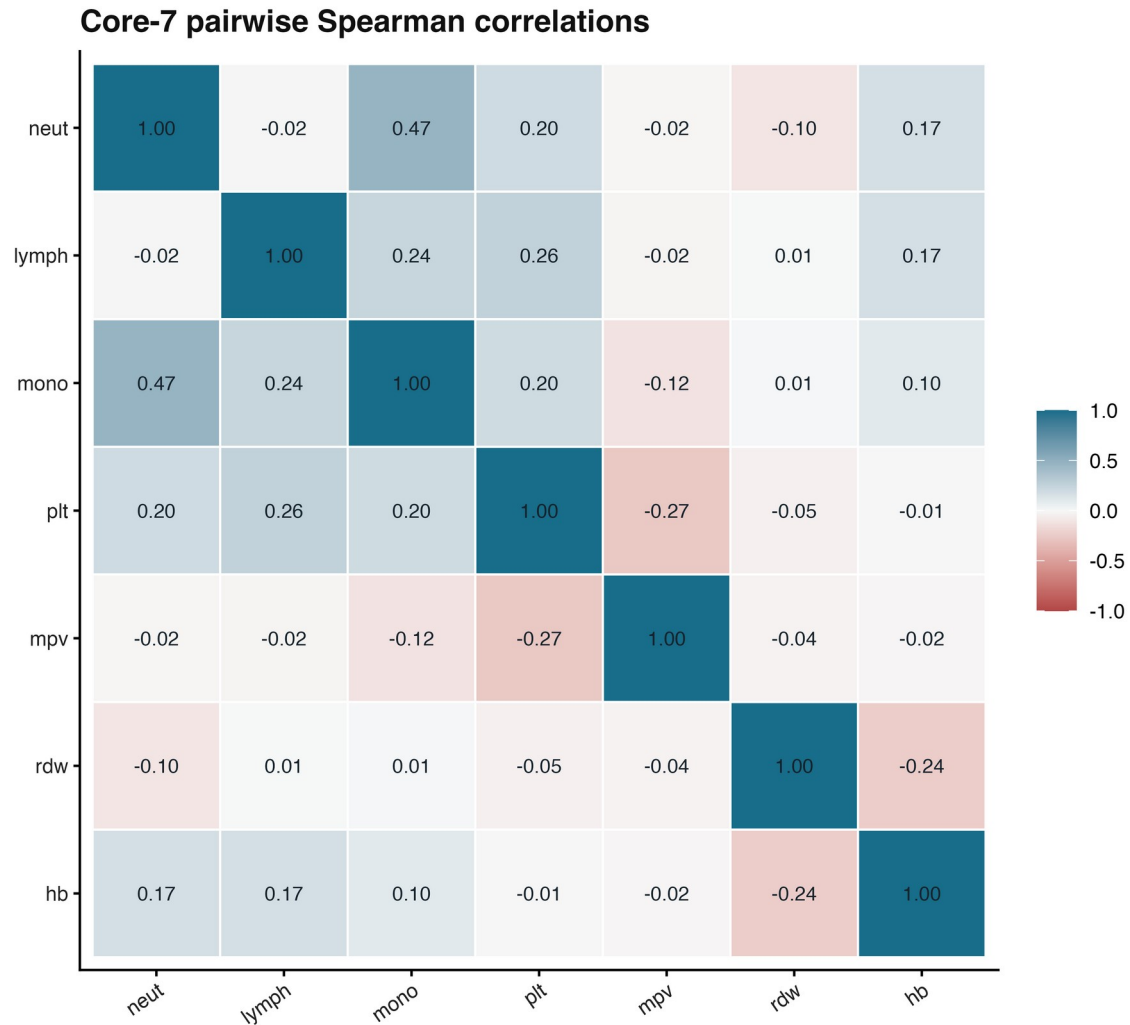
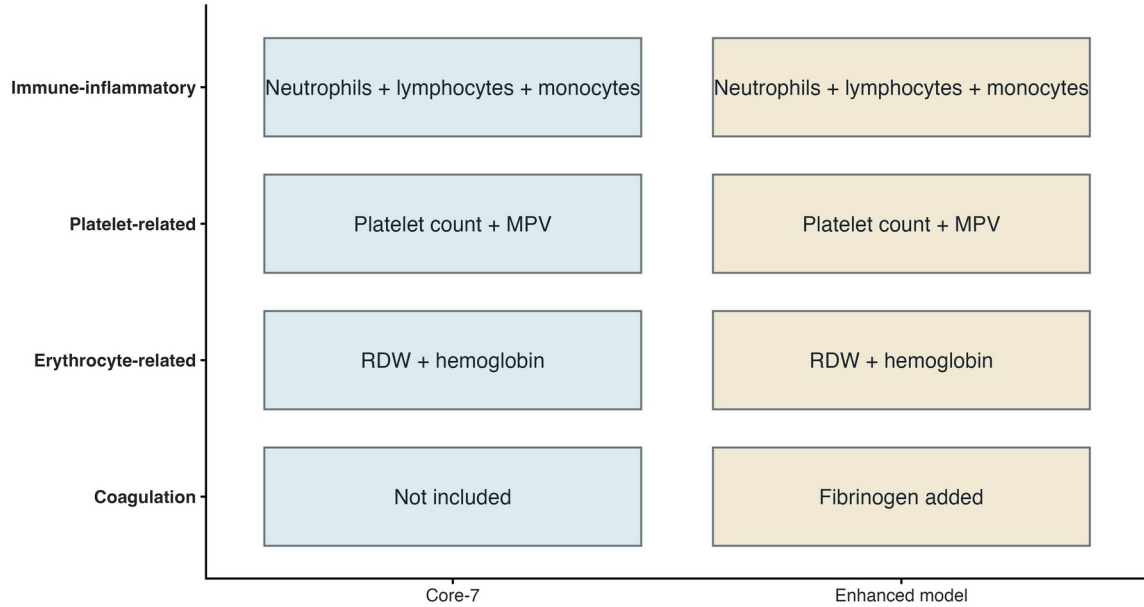


Figure S3. Pairwise Spearman correlation matrix for Core-7 predictors. Maximum absolute correlation was 0.4685 between neutrophils and monocytes.

Analytical grouping of routine hematologic features

Domains organize variables for interpretation; they are not independent causal pathways



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Figure S4. Analytical grouping of immune-inflammatory, platelet-related, erythrocyte-related, and coagulation measurements; groups organize interpretation and do not represent independent causal pathways.

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Supplementary tables

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Supplementary Tables S1-S13 are supplied in Additional file 2 (XLSX).